

Long-Range Off-Grid Connectivity Using Meshtastic & LoRa

Initial Project Report

Prepared by	Christy Sahayaraj
Course	ECE570 Internet of Things
Instructor	Dr. Farid Farahmand
Date	20-September 2026

Note: System demonstrated with two Heltec WiFi LoRa 32 V3 nodes, an Android phone, and a Windows laptop.

Executive Summary

This initial project demonstrates bidirectional, off-grid text messaging using two Heltec WiFi LoRa 32 V3 devices running Meshtastic. One radio node is connected locally to an Android phone over Bluetooth, while the second radio node is connected to a Windows laptop through USB serial. The two Heltec devices exchange messages over LoRa. The system was successfully tested in both directions, including a test with Wi-Fi and mobile data disabled, demonstrating that the inter-node message path does not depend on Internet connectivity.

Meshtastic is an open-source, decentralized mesh communication platform designed for low-power LoRa devices and can operate without cellular towers, Wi-Fi infrastructure, or Internet connectivity. [1]

The initial phase focuses on hardware setup, firmware installation, node configuration, connectivity, message exchange, and troubleshooting. Controlled range and signal-quality measurements are reserved for the next test phase and are therefore not populated in this report.

1. Introduction

Modern messaging normally depends on infrastructure such as cellular base stations, Wi-Fi access points, and Internet services. In remote areas, during infrastructure failures, or for low-power field applications, a direct radio-based communication method can be useful. LoRa provides a long-range, low-data-rate physical-layer radio technology suited to small packets and low-power devices, while Meshtastic adds a practical messaging and mesh-networking layer on supported LoRa hardware.

Meshtastic describes its platform as an open-source, off-grid, decentralized mesh network designed to run on affordable low-power devices. It supports phone applications, a web client, and other interfaces for interacting with radio nodes. [1]

The goal of this project is not to provide Internet access over LoRa. Instead, it is to demonstrate direct text-message delivery between two endpoints using LoRa as the long-range transport, with Bluetooth and USB serving only as local interfaces to the two radio nodes.

2. Project Objectives

- Install Meshtastic firmware on two Heltec WiFi LoRa 32 V3 boards.
- Configure the two devices as identifiable Meshtastic nodes using the same regional and LoRa modem settings.
- Connect Node 1 to an Android phone by Bluetooth and Node 2 to a Windows laptop by USB serial.
- Confirm that the two nodes discover one another over LoRa.
- Demonstrate bidirectional text messaging between the phone and laptop endpoints.
- Confirm that the communication remains functional when Wi-Fi and cellular data are disabled.
- Prepare the system for a later range and signal-quality measurement phase.

3. Fundamental Concepts

3.1 LoRa

LoRa (Long Range) is a spread-spectrum radio modulation designed for robust communication of relatively small amounts of data over long distances. A LoRa link trades high data throughput for sensitivity and range, which makes it well suited to telemetry, short messages, sensor data, and low-power field devices. The technology is not intended to replace Wi-Fi or cellular broadband for high-bandwidth applications.

Several radio parameters influence a LoRa link, including bandwidth, spreading factor, coding rate, transmit power, antenna performance, and the surrounding radio environment. Meshtastic groups many of these parameters into modem presets so that users can choose a practical operating profile without manually configuring every low-level setting.

3.2 Meshtastic

Meshtastic is an open-source communication platform that uses supported LoRa radios for peer-to-peer and mesh messaging. The radio devices can be managed from Android, web, and other clients. Messages can be exchanged without cell towers or Internet service once the nodes and channels are configured. [1]

In this project, Meshtastic provides the firmware on each Heltec device, node identity, channel messaging, device discovery, configuration management, and the client interfaces used on the phone and laptop.

3.3 Nodes and Mesh Networking

A node is one physical Meshtastic radio device. Each node has an identity, user-visible name, radio configuration, and a database of other nodes it has heard. With two radios, the system can communicate directly from one node to the other. With additional nodes, Meshtastic can support message forwarding, creating a mesh in which intermediate nodes may extend coverage beyond a single direct radio link.

3.4 Bluetooth, USB Serial, and COM Ports

Bluetooth is used only for the local connection between the Android phone and Node 1. It does not provide the long-range link. The long-range transmission is performed by the LoRa radios in the Heltec boards.

Node 2 is connected to the Windows laptop by USB. The Heltec V3 integrates a CP2102 USB-to-serial interface, which allows firmware download and serial communication. Windows exposes this interface as a COM port; in the tested setup, Node 2 appeared as COM7. [2]

3.5 Frequency Region and Modem Preset

Meshtastic radio settings must comply with the radio regulations and supported frequency plan of the operating region. Both nodes in this project were configured for the US region. The modem preset used for the initial test was Long Turbo. For two nodes to communicate reliably, their key LoRa settings and channel configuration must be compatible.

3.6 Channels and Encryption

Meshtastic organizes messaging through channels. A channel contains configuration information shared by participating nodes and can include an encryption key. Broadcast messages sent to a common channel can be received by other nodes configured for that channel. The initial project test used the common/default channel configuration so that basic communication could be verified before experimenting with custom private channels.

3.7 RSSI and SNR

Received Signal Strength Indicator (RSSI) is a measure of received radio power, commonly expressed in dBm. Signal-to-Noise Ratio (SNR) expresses the strength of the desired signal relative to background noise, in dB. These measurements are useful for later range testing because they help describe link quality as distance and environmental conditions change. Controlled RSSI/SNR measurements are intentionally left for the next project phase.

4. Hardware and Software

4.1 Heltec WiFi LoRa 32 V3

The Heltec WiFi LoRa 32 V3 is an integrated development board built around an ESP32-S3 microcontroller and an SX1262 LoRa transceiver. It includes Wi-Fi, Bluetooth Low Energy, a 0.96-inch OLED display, USB Type-C, battery-management circuitry, an IPEX/U.FL LoRa antenna connector, and an integrated CP2102 USB-to-serial interface. [2]

Component	Used in Project	Purpose
Heltec WiFi LoRa 32 V3	2 boards	Meshtastic radio nodes

Component	Used in Project	Purpose
ESP32-S3	Integrated	Runs firmware and handles local interfaces
SX1262	Integrated	LoRa radio transceiver
LoRa antennas	2	Transmit and receive LoRa RF signals
OLED display	Integrated	Local status and device information
Android phone	1	Meshtastic mobile client for Node 1
Windows laptop	1	Meshtastic web client for Node 2
USB-C data cable	1+	Power, flashing, and USB serial communication

4.2 Software Tools

- **Meshtastic firmware:** Runs on both Heltec nodes and provides radio messaging/mesh functions.
- **Meshtastic Web Flasher:** Used in a Chromium-based browser to install firmware on the Heltec V3 boards.
- **Meshtastic Android app:** Used to connect to and control Node 1 over Bluetooth.
- **Meshtastic Web Client:** Used to connect to Node 2 from the Windows laptop through USB serial.
- **Silicon Labs CP210x VCP driver:** Allows Windows to recognize the Heltec CP2102 USB-serial interface as a COM port.

5. System Architecture

The final initial-test architecture used different local interfaces at each end but a common LoRa radio link between the two Heltec nodes. This demonstrates that the endpoint client does not need to be identical on both sides.

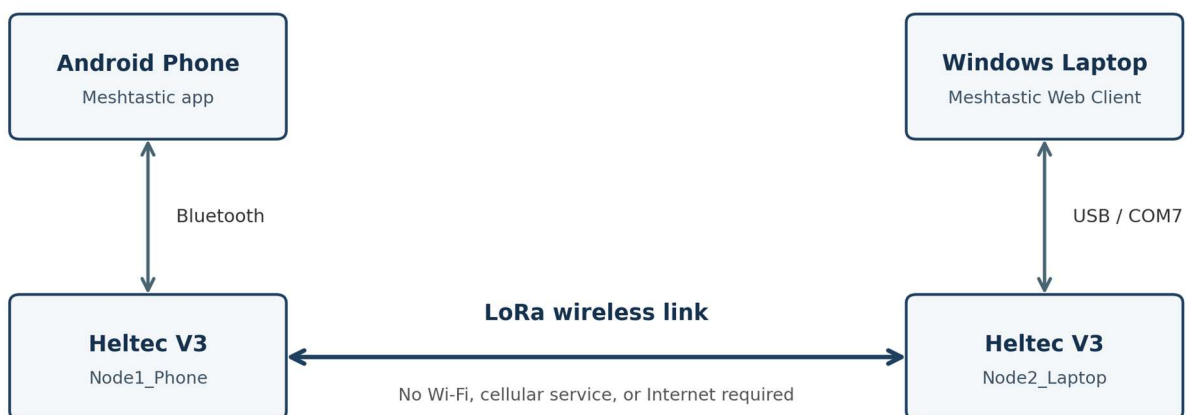


Figure 1. Initial Meshtastic system architecture.

When a user types a message on the Android phone, the message is transferred locally by Bluetooth to Node1_Phone. Node 1 transmits the Meshtastic packet over LoRa. Node2_Laptop receives the packet and forwards it locally through USB serial to the browser-based Meshtastic client on the Windows laptop. The reverse path is used for replies.

6. Implementation and Configuration

6.1 Firmware Installation

Both Heltec boards were connected to the Windows laptop using a USB-C data cable and flashed with Meshtastic firmware using the Meshtastic Web Flasher. The official flasher explicitly requires a data-capable USB cable and supports selecting the Heltec V3 target before firmware installation. [3]

After each flash completed, the board rebooted and the OLED display confirmed that the firmware was running. Both nodes were then kept on the same firmware family for the communication tests.

6.2 Node Identification and Naming

Because both newly flashed nodes initially appeared similar in the Bluetooth scan, one node was temporarily powered off while the other was identified and renamed. Node 1 was assigned the name Node1_Phone, and Node 2 was assigned the name Node2_Laptop. This naming scheme made later testing and node discovery unambiguous.

6.3 Radio Configuration

Parameter	Node 1	Node 2
Device	Heltec WiFi LoRa 32 V3	Heltec WiFi LoRa 32 V3
Node name	Node1_Phone	Node2_Laptop
Region	US	US
Modem preset	Long Turbo	Long Turbo
Local interface	Bluetooth	USB serial / COM7
Client	Android app	Web client

7. Testing and Evaluation

Testing was divided into functional connectivity tests and a later quantitative range/signal test. The functional tests were completed successfully. Quantitative distance and signal-quality values are intentionally reserved for the next measurement session.

7.1 Node Discovery Test

With both nodes powered and configured using compatible LoRa settings, Node2_Laptop appeared in the Android application connected to Node1_Phone. This confirmed that Node 1 was receiving Meshtastic packets transmitted by Node 2 over LoRa.

7.2 Phone-to-Laptop Message Test

A broadcast message was sent from the Android Meshtastic application. The message traveled from the phone to Node 1 over Bluetooth, from Node 1 to Node 2 over LoRa, and from Node 2 to the Windows web client over USB serial. The message was received successfully on the laptop.

7.3 Laptop-to-Phone Message Test

A reply was sent from the Meshtastic web client on the laptop. Node 2 transmitted the message over LoRa to Node 1, and the Android phone received it through the local Bluetooth connection. This confirmed bidirectional communication.

7.4 Offline Communication Test

Wi-Fi was disabled on the Windows laptop. On the Android phone, Wi-Fi and mobile data were disabled while Bluetooth remained enabled. Messages were then sent in both directions. Both messages were successfully received, confirming that the LoRa communication path did not depend on Wi-Fi, cellular data, or Internet connectivity.

Functional Test	Result
Node 1 firmware installation	PASS
Node 2 firmware installation	PASS
Android to Node 1 Bluetooth connection	PASS
Laptop to Node 2 USB serial connection	PASS
LoRa node discovery	PASS
Phone to laptop message delivery	PASS
Laptop to phone message delivery	PASS
Bidirectional messaging with Internet access disabled	PASS

7.5 Range and Signal-Quality Testing

Controlled distance, RSSI, SNR, and message-reliability measurements will be completed during the next range-test session. Quantitative measurement values are intentionally left out of this initial report and will be inserted after the controlled test is performed.

8. Troubleshooting and Lessons Learned

Charge-only USB cable: The first USB-C cable powered the OLED but did not create a serial device in Windows. Replacing it with a known data cable allowed the CP2102 interface to appear. This demonstrated that a powered board does not necessarily indicate a working data connection.

CP2102 driver not installed: Windows initially listed the CP2102 USB to UART Bridge Controller under Other Devices with Code 28. Installing the Silicon Labs CP210x virtual COM port driver allowed the device to appear under Ports (COM & LPT).

Serial port already open: During one flashing attempt, the web flasher reported that the serial port was already open. Closing/restarting the connection and reconnecting the device released the WebSerial port and allowed flashing to continue.

Two identical nodes visible: With both freshly flashed devices powered, the Bluetooth scan showed two similar Meshtastic devices. Temporarily powering off one board made it possible to identify, connect, and rename the first node before configuring the second.

Endpoint flexibility: The second Android phone available for the project used an old Android version, so the design was changed to use a laptop at one endpoint. The successful test showed that Meshtastic nodes can bridge different local client interfaces while the inter-node transport remains LoRa.

9. Discussion

The initial implementation achieved its main functional objective: text messages were exchanged between an Android phone and a Windows laptop through two Heltec LoRa radios, and the communication remained operational when conventional Internet connectivity was removed.

The architecture also clarifies the role of each wireless technology. Bluetooth is only a short-range control/data connection between the phone and its local Heltec radio. USB serial is only a local wired interface between the laptop and its Heltec radio. LoRa is the actual radio technology carrying messages between the two geographically separable nodes.

Although Meshtastic is designed for mesh networking, this initial system uses two nodes and therefore primarily demonstrates direct peer-to-peer LoRa communication within the Meshtastic protocol. Adding one or more strategically placed nodes in a later phase would allow multi-hop behavior and mesh-routing concepts to be evaluated experimentally.

10. Current Limitations

- Controlled long-distance range testing has not yet been completed.
- RSSI and SNR values have not yet been recorded using a repeatable measurement procedure.
- The system currently uses only two radio nodes, so multi-hop relay behavior has not been evaluated.
- Battery runtime and power consumption have not been characterized.
- GPS/location features and sensor telemetry are outside the scope of the initial build.
- Private-channel configuration and encryption-key management have not yet been evaluated as separate experiments.

11. Future Work

- Perform indoor and outdoor range tests at documented distances and environments.
- Record signal-quality metrics such as RSSI and SNR for fresh packets at each test location.
- Compare message success rate and link quality as distance and obstructions increase.
- Add a third Meshtastic node to evaluate relay and multi-hop mesh behavior.
- Operate the Heltec nodes from rechargeable batteries or power banks and measure practical runtime.
- Create and test a private Meshtastic channel with controlled key sharing.
- Investigate GPS/location sharing and sensor telemetry for IoT-oriented applications.
- Explore different modem presets and compare range, responsiveness, and airtime behavior.

12. Conclusion

A functional off-grid messaging prototype was successfully assembled using two Heltec WiFi LoRa 32 V3 nodes running Meshtastic. The first node was connected to an Android phone through Bluetooth, while the second node was connected to a Windows laptop through USB serial. The two radio nodes discovered each other and exchanged text messages in both directions over LoRa.

The most important initial result was successful bidirectional messaging after Wi-Fi and mobile data were disabled. This confirms the intended architecture: local user devices communicate with nearby Heltec nodes, while the Heltec nodes exchange messages directly over LoRa without relying on conventional network infrastructure. The completed setup provides a stable platform for the next phase of the project, which will focus on quantitative range and signal-quality testing and, later, more advanced mesh-networking features.

References

- [1] Meshtastic, "Off-Grid Communication For Everyone," Meshtastic official website. <https://meshtastic.org/> (accessed September 2026).
- [2] Heltec Automation, "WiFi LoRa 32 (V3), ESP32-S3 + SX1262 LoRa Node," product documentation. <https://heltec.org/project/wifi-lora-32-v3/> (accessed September 2026).
- [3] Meshtastic, "Meshtastic Web Flasher," official firmware flashing tool. <https://flasher.meshtastic.org/> (accessed September 2026).

[4] Semtech Corporation, “LoRa Modem Design Guide,” Revision 1, July 2013. Available through Meshtastic documentation assets.

Appendix A. Next Measurement Session Checklist

- Power both nodes and verify Node1_Phone and Node2_Laptop appear in the clients.
- Keep Region and Modem Preset unchanged unless the experiment specifically tests a different preset.
- Choose a documented distance and environment for each test point.
- Send a fresh message immediately before reading signal metrics.
- Record distance, environment, RSSI (if available), SNR, and message delivery result.
- Repeat each distance more than once if time permits to reduce reliance on a single packet.

Project Setup:



Web Meshtastic From Laptop:

	Long Name	Connection	Last Heard	Encryption	SNR	Model	MAC Address
N2L	Node2_Laptop	Unknown	41 seconds ago		0 dB 56%	HELTEC_V3	AC:A7:04:09:09:60
N1P	Node1_Phone	Unknown	2 minutes ago		6.3 dB 81%	HELTEC_V3	AC:A7:04:08:58:F0

Meshtastic Web

client.meshtastic.org/nodes

Meshtastic

Navigation

- Messages
- Map
- Settings
- Nodes (1)

Search nodes...

	Long Name	Connection	Last Heard	Encryption	SNR	Model	MAC Address
N2L	Node2_Laptop	Unknown	38 seconds ago		0 dB 56%	HELTEC_V3	AC:A7:04:09:09:60
N1P	Node1_Phone	Unknown	4 minutes ago		6 dB 80%	HELTEC_V3	AC:A7:04:08:58:F0

Node2_Laptop

- Serial: 10c4:ea60 Connected

97% charging

Voltage: 4.15 V

Firmware: 2.7.26.54e0d8d

Change Color Scheme

Command Menu

Language: English

V0.0.0 - #011EDE9

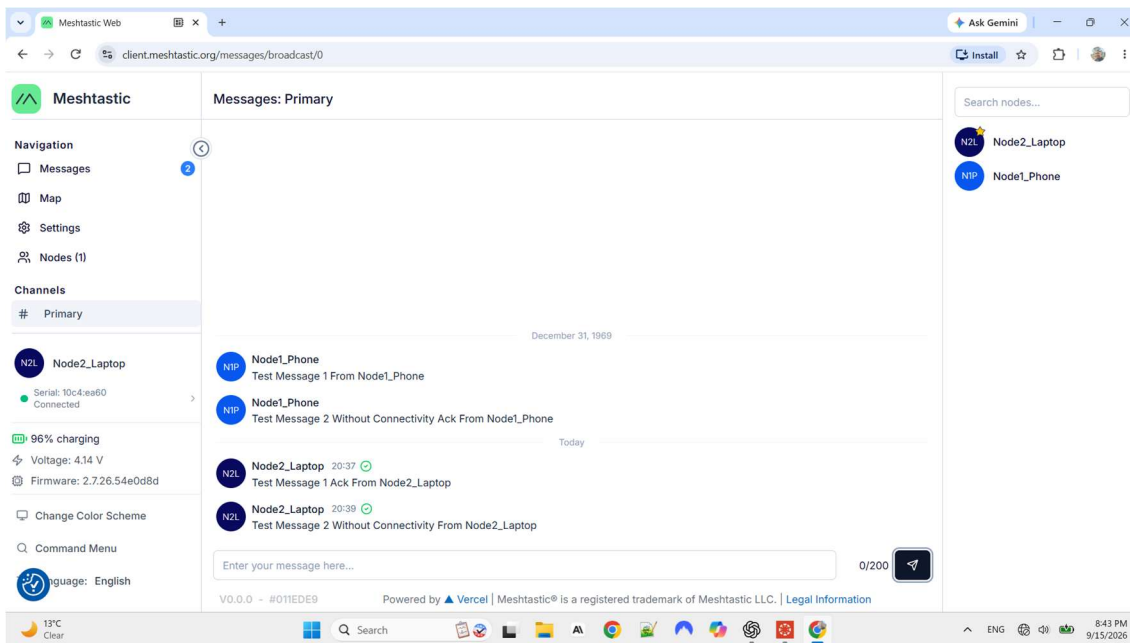
Powered by Vercel | Meshtastic® is a registered trademark of Meshtastic LLC. | Legal Information

13°C Clear

Search

ENG

8:44 PM 9/15/2026



Meshtastic App from Android Mobile:

